NAOMIE BAMBARA

APPLICATION DEVELOPER

800 11th Street South St. Cloud, MN 56301 | (832)-274-4028 | naomiebambara87@gmail.com | www.linkedin.com/in/naomie-bambara-0023201a2 | Projects: https://joliage.github.io/index.html

SUMMARY

Dedicated computer science and global business graduate with experience designing backend systems, integrating APIs, and creating software solutions that support analytics and data workflows. Strong in Python and Java with a focus on functionality, performance, and maintainability.

EDUCATION

Master of Science: **Computer Science** **05/2026**
St. Cloud State University (SCSU) St. Cloud, MN

- GPA: 3.53

Bachelor of Science: **Computer Science and Management: Specialization in Global Business** **12/2023**
St. Cloud State University (SCSU), Herberger Business School St. Cloud, MN

Study Abroad: **Concentration in Global Business and Korean Culture** **12/2021**
Solbridge International School of Business Daejeon, South Korea

SKILLS

- **Machine Learning Frameworks:** Scikit-Learn, Keras, TensorFlow, NumPy, Pytorch, Pandas, NLTK, SpaCy, Hugging Face.
- **Programming Languages:** C/C++, Java, Assembly language, Python, Android Studio, Swift, HTML and CSS, SQL Developer, R Cloud Studio, MATLAB, Jupyter.
- **IDEs:** Visual Studio Code
- **Artificial Intelligence and Analytics Framework**: TensorFlow, Jupyter Notebooks, Google Colab
- **Operating Systems and Environments:** Linux Software Development (Kali, Ubuntu, Git Bash), Windows, Azure Cloud Computing, GitHub, Docker.
- **Relational Database Management Systems:** MySQL, SQL Server, MongoDB
- **Languages:** English, French, Thai.

EXPERIENCE

Database Software Developer, UNESCAP, Bangkok, Thailand **10/2025 to 03/2026**

- Designed and implemented a MongoDB schema to store publications, programs, publishers, and statistics, enabling scalable and flexible data management.
- Developed RESTful APIs using Flask to retrieve and serve structured data to the frontend, enabling dynamic search and analytics features by 40%.
- Established secure database connections using environment variables and python-dotenv, improving maintainability and security of the application by 45%.
- Configured and deployed the application using Docker by setting up containerized environments, ensuring consistent execution across systems by 50%.
- Validated application functionality inside containers, enabling reproducible setup for development and demonstration environments.

Graduate Assistant, **St. Cloud State University**, St. Cloud, MN **08/2025 to 12/2025**

- Dev-ops machine learning analytics practical troubleshooting and implementation neural network concepts, including backpropagation, gradient descent, and supervised classification using Scikit-Learn and Keras.
- Data modeling for integration for the deployment of convolutional Neural Networks (CNNs) and Transfer Learning for medical imaging tasks, such as pneumonia detection in chest X-rays.

Website Designer, **The M.I.S.F.I.T.S Rugby** **09/2023 to 07/2024**

- Enhanced website performance by integrating diverse functionalities, leading to an increase of 40% in user engagement.
- Developed and communicated design plans clearly to stakeholders, ensuring alignment with business goals and user-friendly website interfaces.
- Designed and delivered high-quality WordPress websites with creative and technical expertise, resulting in an improvement of 60% in site navigation.

- Created visually appealing pages using carefully selected color schemes and fonts, boosting website aesthetics and user satisfaction and increased traffic by 30%.

Java Programmer, **UNESCAP**, Bangkok Thailand **01/2023 to 06/2023**
- Developed a Java program for efficient job-to-department matching, optimizing resource allocation processes by 40%.
- Ensured data accuracy verification procedures.

PROJECTS

Text-to-Speech Software in Moore Language Project
- Python development of a lightweight GUI/CLI (Tkinter + CLI) using OOD/OOP scripts approach.
- Python system implementation for automated dataset integration.

Bot Detection & Traffic Filtering System
- Developed a backend system to detect and filter bot-generated traffic by 50% by integrating an external IP intelligence API and analyzing request metadata.
- Implemented API calls using Python (requests) to classify IPs based on VPN, proxy, and TOR indicators, storing results in MongoDB for reuse.
- Built a filtering pipeline to remove bot-generated records from raw analytics data, improving the reliability of download and view statistics by 35%.
- Optimized performance by caching IP classification results, reducing redundant API calls and improving processing efficiency by 20%.

Publication Metadata Ingestion & API Integration
- Built a data ingestion pipeline by consuming 100% of UNESCAP DSpace REST APIs and extracting publication metadata, enabling centralized storage for analytics.
- Transformed raw Dublin Core metadata into normalized application-ready fields using Python, improving query efficiency and frontend usability by 45%.
- Implemented automated data retrieval and parsing workflows using Cloudscraper to handle protected endpoints, ensuring reliable data acquisition by 25%.

Database Design & Backend System (MongoDB + Flask)
- Designed and implemented a MongoDB schema to store publications, programs, publishers, and statistics, enabling scalable and flexible data management.
- Developed RESTful APIs using Flask to retrieve and serve structured data to the front-end, enabling dynamic search and analytics features by 40%.
- Established secure database connections using environment variables and python-dotenv, improving maintainability and security of the application by 45%.

Analytics & Dashboard System (Statistics Processing)
- Built aggregation pipelines in MongoDB to calculate monthly downloads, views, and top-performing publications, enabling data-driven insights by 30%.
- Developed backend API endpoints for dashboards (e.g., trends, top publications, program breakdown) using Python, supporting real-time frontend visualization.
- Implemented dynamic filtering and grouping by program of work and sub-regions, improving analytical depth and user interaction with data by 25%.

Deployment & Containerization (Docker Environment)
- Configured and deployed the application using Docker by setting up containerized environments, ensuring consistent execution across systems by 50%.
- Installed and managed services including MongoDB and backend APIs within containers, simplifying deployment and reducing environment issues by 40%.
- Validated application functionality inside containers, enabling reproducible setup for development and demonstration environments.

CERTIFICATIONS

CAMP – Certified Associate Project Manager **2024 -2027**
PMI Certifications